

Internship Network in the Mathematical Sciences: Year 2 Evaluation Report

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Introduction

The Internship Network in the Mathematical Sciences (Inmas), a three-year project funded by the National Science Foundation (NSF), is co-directed by Fadil Santosa (Johns Hopkins University) and Richard Laugesen (University of Illinois, Urbana-Champaign). The goals of this project are to:

- Provide high quality training and internship experiences that broaden career opportunities for students in the Mathematical Sciences.
- Bring powerful computational and modeling tools to bear on industry's most pressing problems.
- Build mutually beneficial connections between industries and mathematics departments.
- Support systems change within mathematics departments in support of expanded opportunities for students and industry partners.

The major activities planned to accomplish these goals are the following.

- Launch regional networks, with hubs at Johns Hopkins University and the University of Illinois and nodes at graduate level institutions in neighboring states.
- Prepare students for the internships through technical and professional skills training.
- Seek out challenges in industry and government organizations that can be tackled by Mathematical Sciences techniques.
- Create internship projects at those host organizations, and match with students in the network.

In the second year of this program, students selected to participate in Inmas attended three technical skills weekend workshops in Python Programming, Statistical Learning, and Data Science and Machine Learning. Two additional weekend workshops were also included: one focused on resume preparation and the other on teamwork and a group project. In contrast to the first year of the project, which was entirely conducted virtually due to the Covid pandemic, this year's workshops were offered in-person. Two workshops were at the hub locations (Johns Hopkins University and University of Illinois Urbana-Champaign); students gathered at their local node universities for the remaining workshops.

In the second year of the program, 64 of the 111 students who applied to Inmas were accepted. 32 students affiliated with the Midwest hub and 22 students affiliated with the Mid-Atlantic hub completed the technical and professional development training workshops. Forty students (23 Midwest hub and 17 Mid-Atlantic hub students) were placed in Inmas internships. The remaining students applied for other internships or selected to focus on their thesis work.

In selecting companies to participate and host a graduate mathematics intern, Inmas sought companies that had projects involving a high level of intellectual sophistication including some amount of mathematical or statistical reasoning. In particular, Inmas looked for places where mathematically-trained talent could make a difference and where the company did not have a history of engaging mathematical scientists. In the Mid-Atlantic, small-to-medium sized companies were targeted because we wanted to maximize potential impact to the company. The methods by which candidate host companies were identified included:

- recommendations from node university personnel in academic units and corporate relations offices,
- companies with whom the PIs have had contact in previous years,

- companies identified as holding federal Small Business Innovation Research (SBIR) grants, and
- companies located in university-affiliated research parks/business incubators.

Companies that seemed promising were then contacted by email. If they responded positively, a Zoom call was arranged with their staff to understand better their business challenges. Both Midwest and Mid-Atlantic Project Developers helped staff scope suitable projects.

After learning about the company's priorities and the type of skills required for a project, Inmas identified students with relevant interests and skills for the position. Those students' resumes were forwarded to the company for consideration. Inmas staff would start with students from the nearest node university. If no students from that node or from the Math Alliance were suitable for the particular project, Inmas staff would draw on students from further afield in the network.

Evaluation Design

The goal of the evaluation was to track successes and challenges of the project, paying particular attention to any adaptations made based on the previous year, and to provide the program directors with data, observations, and recommendations for continued improvements. In addition to surveying students on their experiences with the five weekend workshops, this year of the evaluation included an expanded examination of the internship experience with more feedback from the participating companies. The following three overarching evaluation questions guided the second-year inquiry.

1. How did students experience the Inmas program activities?
2. To what extent did the technical and professional development trainings affect students' awareness, skills, and confidence?
3. How did students and industry partners experience the internship matching process including preparation, opportunities, and selection?
4. What was the overall experience of the internship including fit of student skill/interest/knowledge with industry and benefits to students and industry partners?

Evaluation Activities

The evaluation activities for the second year included the following:

- Prior to any workshop activities, an online pre-training survey link was emailed to 64 students in October 2021 to learn about their technical and professional skill levels as well as perceptions about industry opportunities. Sixty-two students completed the pre-training survey.
- Following each weekend technical or professional skills training a survey link was emailed to students:
 - October 2021: Python Programming
 - November 2021: Statistical Learning
 - December 2021: Professional Development-Resume Preparation
 - January 2022: Machine Learning
 - February 2022: Group Project
- The February 2022 survey also included a post-training section of questions (to match the pre-training survey taken in October 2021); 55 students completed that survey.

- Following the internship experiences, interviews were conducted in August and September 2022 with a select number of company staff and students. Twenty people participated in the interviews.

Survey data included descriptive statistics as well as qualitative responses to the open-ended questions. All interviews were conducted via Zoom and were recorded with participants' permission. Interview recordings were transcribed and uploaded into NVivo, a program that supports qualitative analysis. Direct quotations from the interviews are used to illustrate thematic points throughout this report. At each round of data collection, findings were shared with Inmas staff to support ongoing program improvements.

Evaluation Findings

The evaluation findings followed by supporting evidence from the data are presented in four areas: technical and professional development workshops, internship experiences, student perceptions of industry careers, and recommendations.

Technical and Professional Development Workshops

As noted previously, three technical workshops and two professional development (resume preparation and teamwork/group project) workshops were conducted during the fall and winter, 2021-22. The technical topics included Python Programming, Statistical Learning, and Machine Learning. Workshops were developed by Inmas staff and TAs, and by externally contracted instructors, building on the experiences of the first year. Each workshop was staffed by 5-6 TAs to support student learning in the fast-paced workshop environment.

Key Findings for Workshops

Student Gains in Technical Skills and Confidence

- Students had varied levels of experience prior to the workshops ranging from little experience to proficiency. Post-training survey responses show a growth in skills in all three areas (Python, statistical methods, and machine learning).

Student Gains in Professional Development

- Students reported an increase in confidence in all four areas of professional development (resume preparation, networking, interviewing, and teamwork).
- Students noted a shift in awareness about how to ask questions or tackle problems in an industry setting.

Student Recommendations

- Most recommendations focused on having more time or structure to learn the technical skills.

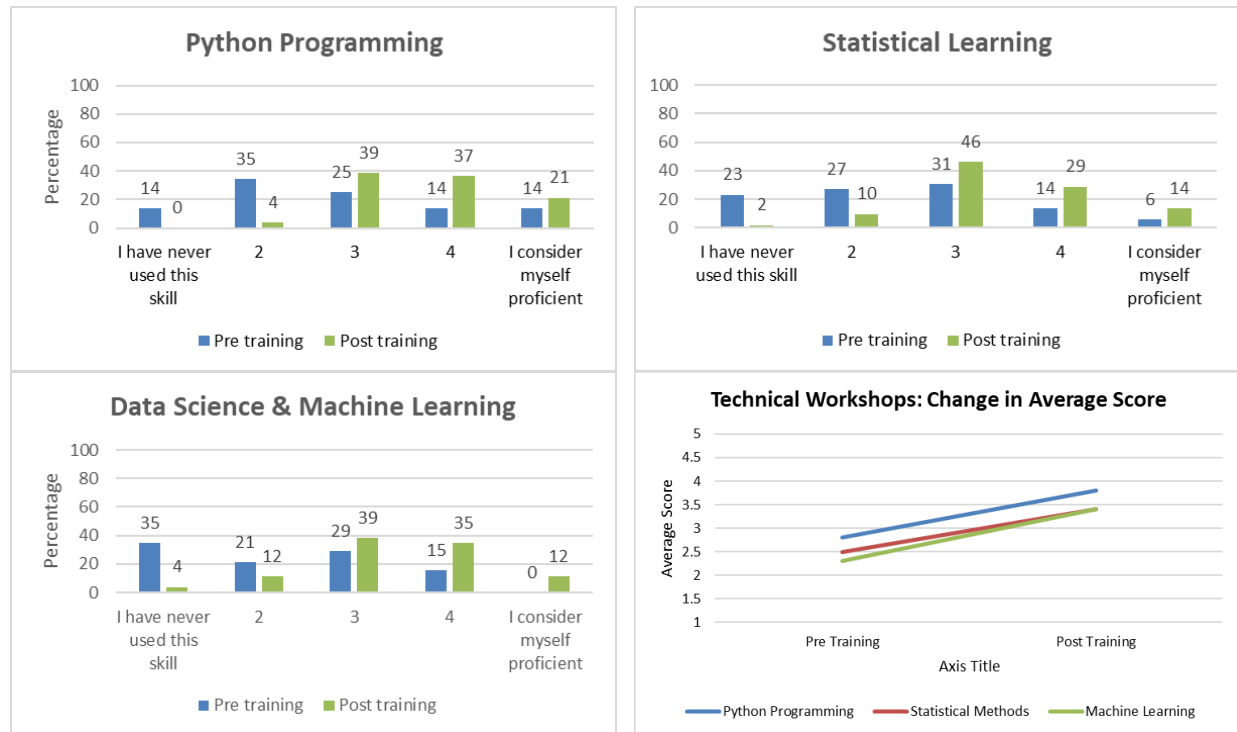
Student Perceptions of Industry Careers

- Overall perceptions about industry careers did not change as much; many questions about working in industry still remained after the workshop.

Supporting Evidence from Pre- and Post-Workshop Surveys

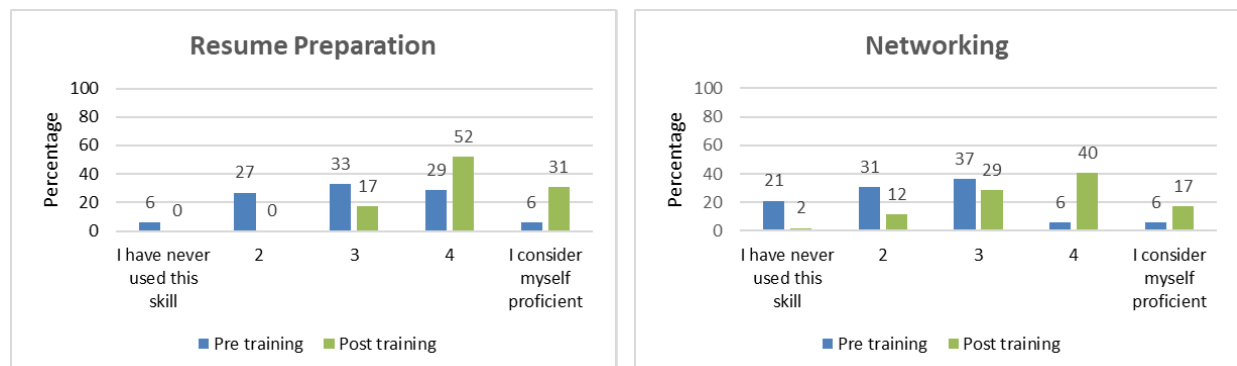
The following are data from the pre- and post-workshop surveys. Invitations were emailed to 64 students; 60 students completed the pre-survey and 55 students completed the post-survey. Students were asked to rate their level of knowledge on a 1-5 scale: I have never used this skill (1) to I am proficient (5).

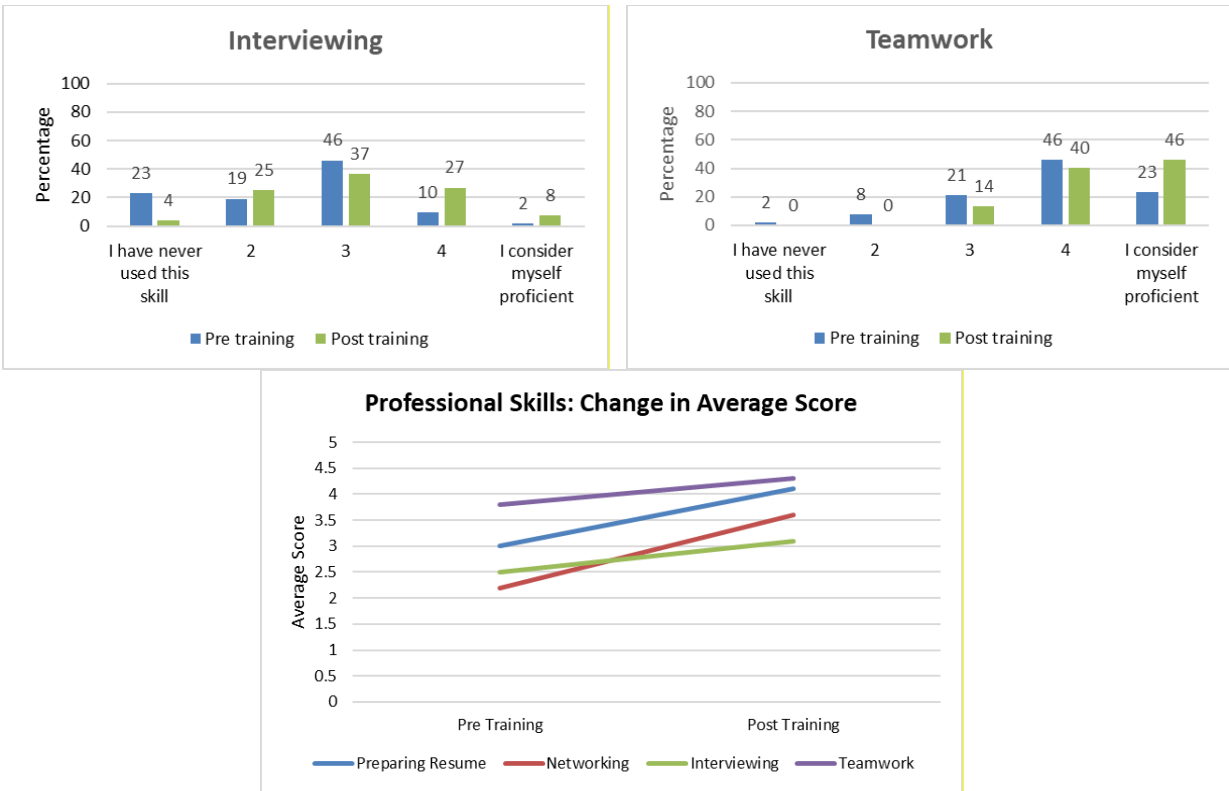
Pre- and Post-Training Survey Results: How would you rate your proficiency in the following technical skills?



Professional development activities focused on resume and interviewing preparation along with teamwork (via group projects). The following are the survey results for these areas.

Pre- and Post-Training Survey Results: How would you rate your proficiency in the following professional skills?





Student Reflections on Workshop Experience

Students were asked what they found most valuable about the workshop experience. The following key examples of their open-ended responses from the surveys.

- I have really found these workshops helpful. A lot of the workbooks were close to the sweet spot of being new/difficult, while being doable. Meeting other (networking with) graduate students from other schools has been a really amazing experience. I really found the workbooks helpful.
- (1) Ability to work in a team: communicate ideas effectively, be able to understand someone else's point of view. (2) Prioritize and deliver under tight deadlines. (3) Look beyond the numbers in the data, make some keen observations, consider corner cases and anomalies, and try to explain them. (4) Network and learn from a community.
- It's important to know what question to ask and (once it's known) to seek existing helpful methodology and materials. Knowing general theory and methods is good, but the thing to keep in mind is where the efforts should be going.
- Working alone can be inefficient, and a team can contribute a lot to your success.
- Being able to know how to ask questions and whom to ask to effectively solve problems is one of the most important skills for succeeding.
- My resume was not anywhere near what it needed to be. I'm glad we had the resume workshops and got to consult with an expert (consultant to Inmas). She is great!

Student Recommendations

Students also shared recommendations for future iterations of the program.

- It might be good to think about whether this can be structured differently. Maybe consecutive weekends, and we have the entire week to work on the notebooks for the training sessions for instance. Right now, we barely get through them and there's no time to get comfortable with any of the material.
- I have one suggestion for the machine learning workshop: maybe we can have two workshops for this topic, one focused on classical machine learning such as linear regression, SVM etc and one focused on deep learning and more latest things.

Internship Experience

As noted earlier, Inmas staff used criteria to select companies with the greatest opportunities to utilize and benefit from interns with high-level mathematical training. In addition, Inmas staff sought to match particular student's interest and skills with the problem identified by companies. These activities were led by Project Developers at each of the two hubs of the project in the Midwest and Mid-Atlantic.

The finding and the evidence presented in the following sections is based on 20 interviews conducted with participating industry partners (16) and student interns (4).

Key Findings For Internship Experience

Inmas Process

- Both industry partners and students reported that the initial phases of the internship process, which included identifying potential companies and projects, assessing student interest and preparation, submitting resumes and interviewing, was conducted well.

Company Characteristics

- Company characteristics such as previous experience with interns and motivation for hosting an intern varied widely. These two characteristics did not appear to determine quality of internship experience.
- Strong connections with a technical mentor/supervisor or team along with structured activities geared toward the internship experience had the greatest impact on the quality of the internship experience.

Overall Internship Outcomes

- Students reported that the internship experience expanded their awareness of industry career opportunities and helped to clarify individual interests.
- All students reported that they were satisfied with the final product of their internship.
- All industry partners reported that they were satisfied or extremely satisfied with the internship outcome.

Recommendations

- Companies were interested in becoming more of a partner with Inmas and the intern in supporting a successful internship experience. For example, companies recommended that Inmas share project reporting requirements with the industry partner.
- Industry partners observed the tensions between solving problems and needing to meet deadlines and messy data sets when working in industry positions. They highlighted the need for students to be aware of this difference with academia. Additionally, the need for being self-directed while also communicating clearly on a regular basis.

Supporting Evidence from Industry Partner and Student Interviews

Student Feedback on Inmas Process

Overall, students who were interviewed reported satisfaction with the process. One person noted that he felt his lack of applied experience was a deterrent.

“I think, all things considered it went really smooth. At one point, Inmas staff met with us and asked all of us about our interests. And after that, he (Project Developer) just compiles a list of companies and he says, ‘How do these sound?’”

“I was really struck by how straightforward it was. I think the thing that stuck out to me most was the fact that, okay, I sent my resume off and all of a sudden here come interviews, and that was it. That was a lot of value right there of not having to spend my time doing that, I guess. It (the selection of internships) was pretty customized because Inmas staff met with all of us for a short 15 minute meeting and asked us what our preferences were. I told him, no finance, no oil and gas, no big tobacco. I'd prefer to do something more social oriented. And so he came back with three different internships for me.”

“Inmas staff started sending out my resume to different companies, but those companies focused on data science and the machine learning aspects of this program. My background in those areas are actually very weak. So those companies didn't really want me in the first round of interviews. But with the company I interviewed with, it was different. They wanted someone who understands math at the college level, at least. And then they wanted someone to build a YouTube channel. The project went really well in my opinion.”

Industry Partner Feedback on Inmas Process

All of the industry partners who participated in interviews said the process of communicating with Inmas and selecting an intern for the summer was smooth and straightforward. Sometimes the company was interested in shaping a project to fit a student's interests and skills; other times the company had a specific project that required a student with particular skills. In either case, industry partners noted how responsive Inmas Project Developers were for their needs.

“Most of them (student applicants) were pretty impressive from their CV, from their resume. I never had a feeling they weren't qualified. So it was basically more that I looked at the candidate, what they were doing or what their expertise was, and then I kind of tailored the project towards their capabilities and what they could do. ... I mean, we were not really looking for an employee. It was more the other way around a little, that we tailored the project more towards.”

Industry partners also reported that they were pleased with the number and quality of candidates that were recommended by Inmas staff.

“So for me, having actually a more curated smaller number of resumes was helpful.”

“I could tell somebody took the time to make sure that the applicants were within the scope of what we were trying to do, so that made our job a lot easier. And then it also was a great use of our time.”

In a few interviews, industry partners noted that they felt they were late in starting the process of selecting interns. Sometimes that meant there was not enough time to extensively review resumes or that candidates had already selected other positions. However, those companies still reported that they were pleased with the intern who ultimately worked with them.

“We got to the place where we were ready to explore applicants a bit late in the season. So a lot of applicants had already found spots. But in spite of that, I mean we still were able to review, I don't know, maybe eight different CVs. And within those were, I would say three really strong candidates.”

They also noted that the entire process—including paperwork—was easier than they had expected. The partners who participated in the interviews were evenly divided in their assessment of the importance of the NSF funding for the position. For some companies, especially the smaller start-ups, the additional funding allowed them to offer an intern a competitive salary. For others, it did not figure into the decision to select an Inmas intern.

Previous Company Experience with Interns

Companies varied in their previous experience hosting interns. Three of the companies interviewed had little or no previous experience. “I've never had an intern before. So it was super, super cool. Now I'm like, I want an intern every year. This is really cool.” Others noted that their company had extensive experience with interns; however, for many of those, this was the first time working with graduate-level math students. More typically, they worked with business or software development students. Two companies interviewed had hosted Inmas interns in the previous year. One company hosted a number of interns: “We had six total this summer. Four of them were engineers and two from math departments, one hosted by Inmas.”

While some of the companies that had extensive experience also had more established structures – such as who to report to or additional meetings geared towards interns—experience alone did not necessarily affect the quality of the internship experience. One internship that was a reported stellar success was with a company that had never hosted an internship before.

Motivation for Hosting Interns

Companies also varied in their motivations for hosting interns from needing extra staffing to identifying and vetting potential employees. Often those motivations overlapped.

“And of course, we also saw a great opportunity in that because we're always short on people. We need help, always help since we are a small startup. And I think students, graduate students, especially if they're just coming from the university, their mind is fresh, they're not biased, and hopefully they're also ambitious and curious. So they're actually the perfect candidates for working with us.”

“We are interested in basically keeping that student engaged in our work and hopefully we can also hire that student next year depending on budget. So that basically, just from that example, the Inmas program was for us actually very successful. I mean, we found someone who could become a potential employee. He was the right fit in terms of qualification and also in terms of personality. ... For us actually to find someone we could employ later on long term, that was more important than the actual work he did this summer.”

“We are a startup company and sometimes as a startup, it's a little hard to hire people, so one of the motivations is that it generates a pipeline of people to possibly enter the company after the internship is over. And as a startup, you continually have to generate talk about the company or else it just disappears into nothingness. So it helps do that-- interns go back and they talk about the internship with other people in departments at school, and it helps generate more people who could possibly work for the company as well. And then pragmatically, there are some tasks that, as a startup, we need some help with, and interns help with that as well.”

“It was very helpful, in fact, to have had the extra \$5,000 matching from NSF. Because I think without that, if it would've been him against some of the undergraduates. I think maybe I would've hired one of the undergraduates, frankly. You've got to balance mathematical ability with communication ability because both are important.”

“But I also wanted to acknowledge that while we were benefiting from this opportunity, I wanted an intern joining us to also benefit. And so wasn't necessarily looking for someone who had 15 years of corporate experience. I was just looking for someone who was excited to dip their toe into a business experience.”

Mentorship and Team Culture

One of the common characteristics of internships that were highly rated by both the industry partners and the student interns was the level of connection between the intern and others in the company. These connections might be with just the company mentor or supervisor or with teams overall. The factors that contributed to these connections included (1) company organization and structures (i.e., teams and intern-specific activities), (2) a technical key contact or supervisor/mentor, and (3) the social and professional skills of intern. While every person interviewed noted success in project completion within each internship, higher levels of connection resulted in more satisfaction.

Student Intern Perception of Mentorship/Team Culture

The following are four descriptions from interns about their experiences. The first two are descriptions of companies with more structured mentoring or team interactions.

“I also said to my mentor (at the company) at the end of the internship, ‘I don't know if you guys were told how to set this up for us or if it was something internal.’ Because I thought it was pretty spot-on how they mentored us through the internship. We had 30-minute meetings with our mentor three times a week, Monday, Wednesday, Friday. And those were just check-in times to be like, ‘What are you working on? How's it going? Do you have any questions that you're getting hung up on?’ So it was just such a good pace of checking in. And then there was a reading group that happened on Tuesdays as well. It was really cool; it felt a little academic. We were just reading research papers as a group. ... I really hope other Inmas people had something similar to that happening in their internship.”

“I thought it was really cool that the company had a whole team that was focused on this stuff, so I could ask anybody questions and they were all on Teams. If I ever needed anything they would just be like, ‘Yeah, I can hop on a call and show you how to do this.’ So everybody was really helpful. I think that was the first thing I noticed straight away was that you never really got the sense that people were too busy for you, which was nice.”

"I was a one man show mostly. So I didn't really work with anyone. I had to report to my manager at three o'clock in the afternoon every day. But besides that, I didn't really seek any help from anybody else."

"It was stressful, actually. I enjoyed the work, but I've never been in a setting like that before. My supervisor, who was the CEO, she was always very, very helpful with anything that I needed that was not technical, right? But at the beginning, I expected them to give me the data that I was going to work with. I tried to set meetings with the right people, and things just kept getting pushed back. And I was just very stressed out like, 'Well, who is my go-to person in terms of the actual work that I do?' That took some time. ... So, I can't stress enough how stressful it was just not knowing exactly what to do, because time is very, very short in the summer, and I need to deliver something. It took me a while for me to shape what that something was. ... I got a lot of help from Inmas Project Developer, actually. I reached out to him and I said, 'How do I handle the situation?'"

Industry Partners' Perception of Mentorship/Team Culture

The following is one description of mentoring from an industry partner's perspective.

"I managed the intern directly over the summer. We had our one-on-ones together where we asked him, 'Okay, what are you trying to get out this summer?' One of his goals was he wanted to get better at presenting. I was like, 'Awesome. We're going to have you present your work as you go and we'll have people from different departments show up who have no idea what you're working on.' So you can explain it from soup to nuts. Then also bringing the intern into our technical team to meet with engineers. He ended up doing his presentation for our C-suite at the end of the summer."

Student Skills and Preparation

Student Perception of Their Skills and Preparation

In general (based on the limited student interviews conducted along with perceptions of industry partners), the internship projects often involved building on mathematical knowledge to varying degrees while exploring new areas such as programming or statistics. In some cases, the internship was quite different from the student's graduate study, and in one case at least, the tasks were well below the capabilities of the student.

"I know some other people in my department ended up finding internships at various national labs and stuff where they were directly using their topology background or something. So I was aware that I could have done that, but it just wasn't something that I was looking for all that directly. For me, it was more about trying to pick up a new skill of data science stuff that I don't do in my day-to-day life, and then applying that to something good for the world."

"No, the math required is actually only in the college level. So there were three main courses that I had to go through to select to put on the YouTube channel. And those three courses are really just college level math. So I would say even under the sophomore, or about the sophomore level. Calculus, linear algebra, differential equations, and other seminars that are also covered in those three courses. So yeah, I would say that any sophomore, not necessarily in math, in a quantitative field, could have done this project."

Industry Partner Perception of Students' Skills and Preparation

The following descriptions illustrate the assessment of student skills, both technically and professionally.

"I would say his level of preparation on the data side was very strong. He was able to come in, start working with data and clean these. And again, these data sets are extremely large, hundreds of thousands of people with maybe, I don't know, a hundred different variables. So these are very large, very complex data sets that would be impossible to work with in something like SPSS."

"We needed hard skills, math, but also really strong programming. There's a lot of matrix math and getting into deep learning. So deep learning and then programming and Python were the major hard skills. Soft skills, though, there's a lot of independence needed. Sort of everyone's busy. Here's something to work on. We had a mentor to work with, but he was asked to really figure a lot of things out on his own and really reach out if he needed help. And so we looked for that as a level of independence and ability to speak up if there's a problem and not wait for someone to find out that you're struggling. So that was really important too. The intern did a great job there. Approached the problem in a few different ways. Talked to people and was proactive about setting things up. Did a lot of work on his own, but he documented it pretty well, and just really, really great job."

"Another thing I really appreciate about the intern is he was great at raising his hand right away and saying, 'I actually don't have a lot of background in this.' Early on he said, 'I understand the math, but I'm not super fluent in Python or the ability to translate this. I can calculate today's data set, but you need something that's going to be able to take data that's coming in and continue to spit out an analysis.' Our CTO jumped in a couple of times with resources. And then the intern was really, really self-led. I know he was reaching out to his advisors, reaching out to classmates and other people that he just knew in his network. Then he would come back to me and say like, 'Okay. Here's what I'm thinking. I learned how to do this thing. I still don't feel like I'm absolutely fluent in it. But if someone is willing to check my work from the Python side of things, I feel like I can still move this forward enough. So he was looking at both internal and external support for reviewing some of the things that he was doing. He's great at research."

"Our intern was definitely fairly independent on the programming side of things. There were of course bugs and issues that we encountered all the time, but I think that has more to do with coding in general than a workshop can't prepare you for every potential problem."

"He really exceeded our expectations. Luckily, we started off on the right foot where we really took the time to talk about what we're doing, what our company's about, where we're going. I could tell he was excited about affecting change. He thought about things and he followed up with questions. He had some really, really good questions. Sometimes interns are a little gun-shy to ask questions. But the problem is, if you don't ask questions then you don't get down to the point you really need to get to. And he had some great questions and we were able to, I think, answer them correctly, because everything he put together was pretty freaking awesome. He was just a really cool professional kid."

"I was more comfortable at least, with giving him a whole lot of freedom in terms of what he did. In terms of the best way to accomplish that nebulous goal I gave for him. He went through,

and he read several different papers. And he came back, and he reported to me about what he thought he would like to do. And the technique and the approach he wanted to take. And we had some good discussions about that. So he is very autonomous, right from the start. I thought he had a great foundational understanding of everything that he was working on. And like I said, I did start him off by doing a little bit of research. I can't tell how much of that came from the time he was spending reading papers, and trying to understand what would be the best approach versus the training that you provide versus the training he had had previously."

"So the first project was really easy, really smooth. We got going and I think all together, we really kind of rocked it. It worked because the project description was something that was directly translatable to the person. And we decided to do it in two phases. The first phase I essentially said, 'What's your favorite tool that you... what's your playground? MATLAB, Mathematica, whatever. Do something you're familiar with and let's just create a mock-up. Let's just figure out the basic nuts and bolts, the basic structure of the problem we want to solve, but let's do an environment that's familiar to you.' So we did that pretty quickly in three or four weeks. And then, we said, "Okay. Now let's move this to our environment," which is we do stuff in the Python programming language. She knew really basic Python, but it was fine. And that's why I paired her up with a software developer so that he could create the infrastructure and guide her along. She was in a pretty kind of self-contained of a person and that worked really well. So her background skills were totally fine. She didn't pretend to be an expert Python programmer. We didn't expect her to be an expert Python programmer. It worked perfectly fine. The second project was trickier because we wanted to do was start with an open-source piece of software, and there's always issues in installing in open-source pieces of software. My team has a lot of experience in some programming languages, but this was a new thing that none of us had really installed. So we essentially underestimated how hard it would be for the intern to just take that piece of software and install it. It didn't look like a huge thing by itself, but it's something he had never done. It's something we hadn't done that exact thing already. So we couldn't just say, 'Oh, yeah, you need to do this and this.' So it took a lot of figuring out just to get up and running, but we got there. But I think hopefully it was a good learning experience. We've learned a lot in terms of what that tool could do, how we needed to adapt it. So it was definitely a good learning experience for everybody."

"We had three interns. The INMAS one was probably the most professional. I mean, everything down to he was well dressed and he was always there on time, really polite and everything. But I think another important thing, especially in a big organization, is just being able to assert yourself a little bit and really pull the information or set up time with people and proactively go and do some of the networking and things. And he was pretty good at that too. He would try and reach out and talk to people, which was good."

Company Interest in Future Participation

All of the industry partners who participated in the interviews indicated that they were interested in participating in the Inmas program again. The only caveat was that it would depend on what was happening in the company, in terms of available resources and time as well as wanting to have a great project for an intern to work on.

Student Perception of Industry Careers

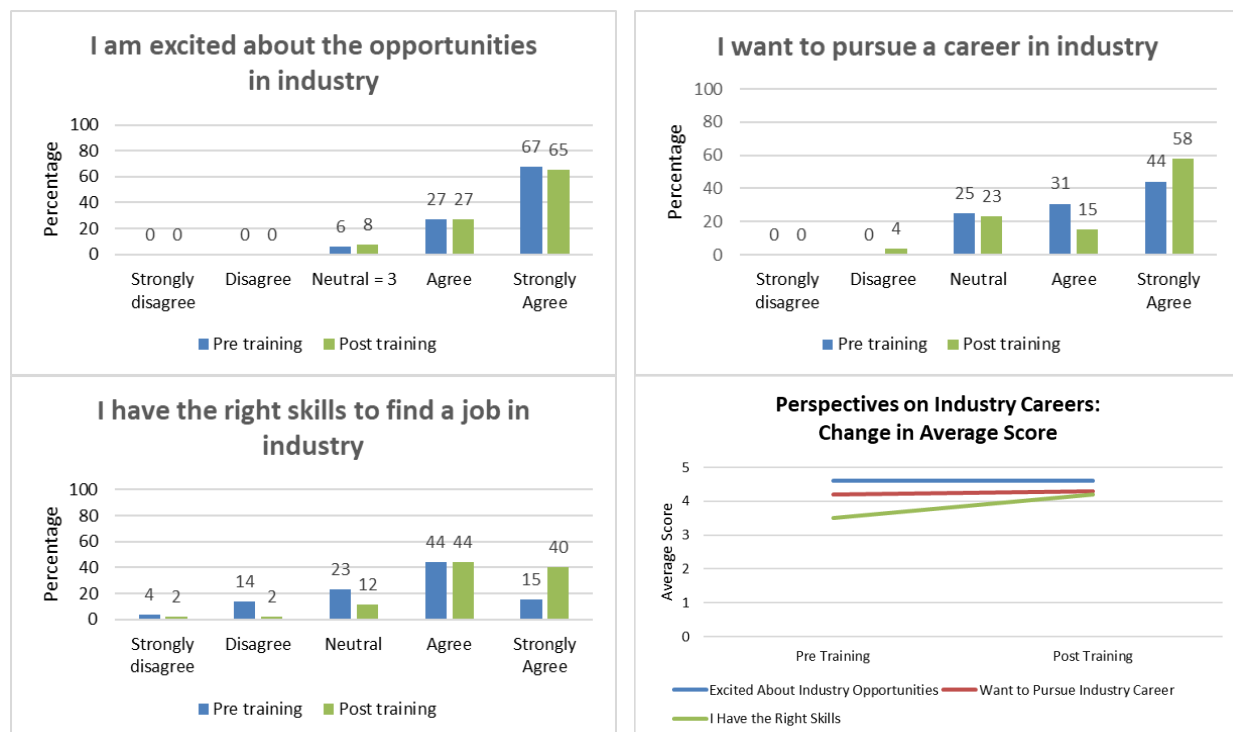
In addition to increased knowledge and confidence in the technical and professional skills discussed earlier, a key outcome the program sought to achieve was an increase in students' awareness and interest in industry career opportunities.

Key Findings Regarding Student Perception

- The data obtained via the pre- and post-training surveys indicate that the weekend workshops did not significantly change students' interest or awareness; however, the interviews that were conducted following the internship experience revealed a much greater shift in perception and excitement in industry positions for many students.

Supporting Evidence from Pre- and Post-Workshop Surveys

At the end of the five weekend workshops, students still had questions about how well their skills and knowledge fit industry opportunities. The following are from the pre- and post-survey along with representative open-ended responses students gave on the post-workshop survey (the survey was administered after the last workshop but before any internship experiences).



- It still remains mysterious to me because I have never had an internship. One of my biggest questions is whether I will be using any of the mathematics that I had to learn in graduate school.
- How do I prepare for technical interviews? Will anyone be interested in hiring me?
- How to find a job in industry and explore the types of jobs available? I want to know which position is a good fit for me.

- My biggest worry is that there might be more to data science that would be expected of me than we learned in the workshops and I would just be oblivious. I also worry that, although I know how to write and run the code in a Jupyter notebook, I don't know how to implement it in a larger program. Sure, I could put together reports, but I wouldn't know how to write a program that would automatically analyze data, I don't know how to put together classes or functions with my data science skills, and it seems unrealistic to do all this "by hand" every time in industry.
- Do a lot of companies make use of teams for projects or is it an individual effort that partners with others?
- I probably have some questions about my skills in NLP and image processing. They seem quite necessary now I'm industry but I didn't understand everything we covered in the training.
- How to efficiently work and communicate with people from different subjects, to come up with plans combining all talents while sufficiently utilizing and showcasing my own expertise.
- How forgiving will an employer be about me learning on the job?
- How can I make the most use of my PhD? While this was fun, I feel like I could have gone into ML immediately after my undergrad and had better prospects than I do now. Cutting loose the sunk costs, how do I recognize and use the aspects of my PhD program that weren't wasted?
- What more can I do to seamlessly transition from being a PhD student to working in industry?

Supporting Evidence from Post-Internship Interviews

As noted above, students described a larger shift in awareness about industry careers after the internship. While that might be self-evident—direct experience often leads to more awareness and understanding—even those students who had challenges in their internship still described the experience as relevant and useful to their career explorations. For example, a student who was returning to an academic path after the summer viewed his internship experience as useful in the ways it would inform his future teaching. However, it also noted a tension with the opportunity costs of now being behind in his research.

Student Perceptions on Internship Outcomes

"I think this was probably the big outcome of the internship for me, was just realizing that I could be really happy in academia, and I could be really happy in industry. Before this, I was like, 'Oh, I don't know if I see myself as a technical person. I love pure math, but if I weren't going to do that, I think I would need a total career change. I don't know if I like computers, I like ideas sort of thing.' And then doing this showed me that, oh, they're actually way more related than I thought and I also feel like I can do it. I wasn't quite sure that I had those skills, and so I was like, oh, that was intellectually fulfilling. It was challenging but doable, and it was towards some purpose that I think is worthwhile. Before the internship, I was like, 'Okay, if the whole mathematician thing doesn't work out. I'm going to have to go into journalism or social work or something else.' And then after, I was like, 'Oh, this is actually pretty cool.' So it just showed me that not only is that possible, there's also just tons of jobs out there that satisfy those components."

"I think the complete answer to the question would be, starting chronologically, the project that I was hired for is not directly related to teaching, but I am very interested in teaching and I said I

thought there would be connections there, during the interview, and they hired me. Coming away from the project, I think that was true. I think a lot of what I learned is relevant to pedagogy and in that sense I think I'm taking things back to my career that are going to be useful. ... But as a matter of research, it's also somewhat connected to projects that I did have some interest in as a hobbyist more than as a mathematician. But I am going to a math and statistics department and so I imagine the students I'll be working with will have some interest in statistics, and I imagine that students I'll be working with wherever I end up, if it's an academic position, will probably be interested in the types of things that I was doing. I think that this experience is going to probably make me a better teacher and make me a better resource to students. However, there's so much pressure to produce research that I think that actually for me, the experience was not worth the opportunity cost of being able to spend that time on research. Now that it's clearer that I have a better chance at having an academic career, which is a hard thing to do in math and I sort of didn't think that it was going to pan out. Now it feels like there's a shot again. Yeah, the opportunity cost I think was a little too high. It was a lot of time and energy spent not doing my own research and now I feel really in trouble as I'm going into this semester. Like, "Oh, I have so much catch up to do."

"It was a great experience. What I think I got most out of it was that I am still okay with working at a startup or doing another internship at a startup. I just don't want to be considered the expert and being a one-man team. That is stressful. I would like to be in another place where they have maybe, sure, maybe they have a research and development section or team, but somewhere where I can refer to or get guidance from, rather than just, "You're the expert. You figure everything out." That was stressful."

Industry Partners' Perceptions on Internship Outcomes

Industry partners were all satisfied by their experience with Inmas interns (evidenced by the descriptions of interns' skills above and additional quotes in the following section).

"I have to give the intern credit because one of the things we did with our data set is to encrypt all of the identifiers. But when we did that, we didn't really foresee that when you encrypt, let's say like an account number or user id, you're going to have a string of numbers that's maybe 15, 16 characters long. And what Excel and even certain statistical packages love to do is to round those numbers up and to treat them as if there's scalar variables instead categorical. And so he discovered we had lots and lots and lots of faux duplicates. They looked like duplicate data because of that rounding error. And so he identified that, fixed that, and then we communicated that out to some of our other research partners who then identified it and fixed it. So that was a nice win on our side. He had extreme attention to detail. We'll miss him when he leaves in December for sure."

"He (the intern) was on the fence as to, 'Well, should I go into academia after I graduate or should I go into industry?' And that this was a great program for him in that he had no idea what industry was. So him being able to spend a summer with us when I asked him in the exit interview process, what did he get out of it? It was that he's much more comfortable and not as scared of what industry would be like. Yes, he could work in either side now, whereas before he knew what academia was like, but that other side was unknown."

"So the intern also participated with that with our entire clinical operations team. And they thought he was just absolutely magical, which was also really enduring and really cool to see him get all of this clout."

“I personally like to work with people who can solve very difficult mathematical problems because that's what we also need. So for us, it actually works pretty well if someone is a really, really good mathematician and knows all the stuff. ... if the person has an interest in problem solving, I think it doesn't really matter much if it's a pure mathematical problem or if it's a problem, it's more of an applied feature. I mean, I don't think so that mathematicians have it more difficult to enter working in an industry or for a company.”

Recommendations from Industry Partners

Industry partners noted a tension between solving problems, or researching an issue, and needing to meet deadlines or communicate to others in the organization. Additionally, they noted that data sets in ‘real life’ are messy and require significant work to structure and clean before any analysis can occur. They wondered if these issues could be addressed in some of the trainings offered by Inmas or even within the departments or acknowledged as part of the learning curve of an internship.

“Oftentimes in industry you're trying to figure out how much do you ask for help. When do you ask for help versus ‘Hey, I don't want to bother Bill.’ The right response will be hopefully politely, ‘Go Google. Have you Googled that yet?’ Or, ‘Go talk to someone.’ So figuring out what that flow is or what the right level of autonomy and I'm going to research this myself is. But at the same time if you're working on your PhD and you spend a month working on something and it doesn't work out, then that's probably totally fine. That's an expected process of growing and learning how to be a researcher. But if you're getting paid a lot at a company and you don't do anything for month, that's not good, in most businesses. So that kind of experience, I think as much as the outcome that the intern had was great, but I think for him, the benefit of getting that comfort level of seeing how I'd interact in an office environment or a corporate environment as opposed to academia was good.”

“I didn't really see this as much with our intern, but I think a lot of times, in an academic environment, a lot of the problems are prepared for you and there's maybe an answer you actually get to. In real life, you don't necessarily get to that. We deal with a lot of messy data, messy configuration issues, things like that. So I think working in the trenches is something that not just for the Inmas interns, but other candidates that come through. They're just like, ‘Oh. There's a lot of other things I have to do.’”

“But working in a company, especially in a small startup, communication is very important. When there's a problem or something, they hit a wall, it's also important to not spend too much time on it. ... So I think one part is planning, work planning, that they need to be transparent. If you have an idea and you want to work on a problem, it's also important that they communicated how they want to solve it, that's their way so that they get feedback from early on that what they're doing is the right way of doing it. Or you can tell them, ‘Hey, listen, forget about it. You will never get anywhere with it. It takes too much time.’ So it's kind of this planning and being transparent about it and then also later on, it's about reporting. They don't need to report only on the good things they have accomplished, also on all the problems they faced because other people, they would like to know that they want to avoid making the same mistakes. It's also very important to meet a deadline. You don't need to be perfect, but you need to meet a deadline or at least you need to provide feedback to everybody else who was involved that you know where they are.”

Industry Partners were also interested in being more involved in the project report requirement for two main reasons: (1) To provide early guidance about handling proprietary information and (2) The project report is also beneficial to the company.

“I already had a structure in my head of like, ‘Oh, the intern is going to have these one-on-ones and he is going to have these touchpoints, and we're going to have this kind of structure around it.’ I was kind of curious at the beginning, wondering, ‘Is there anything I'm supposed to be doing besides that kind of stuff in my own head?’ And so Joe had mentioned to me that he had a report at the end of the summer. So there was something that we were actually working on together where I was like, ‘Okay. Let's make sure that the things that you're working on are documented in a way that we're moving towards this kind of end goal.’ But I wasn't exactly sure like was there anything else specifically that we should be doing? So maybe even just a one pager of like, ‘This is what Inmas wants to get out of. These are the core competencies that we're hoping that students come out with or the skills that they get exposure to or the experiences that they get exposure to.’ Any sort of standardization or just high-level goals, I think would be useful.”